Credit Risk Portfolio Monitor: architecture

Stages 1 to 5 run once, as a batch: python -m src.build. The hosted app reads their outputs. The only call to an outside service is to Gemini, and only when an analyst asks a question.

1. Raw data

5 CSVs in **data/raw/**, unchanged: customers, loan_accounts, bureau_credit, repayment_history, collections_events.

Tool: pandas

2. Ingestion

src/ingest.py, features.py

Checks keys, cleans, logs every decision, joins to one row per account. Bureau data is limited by date to avoid look-ahead.

Tools: pandas, SciPy, PyArrow

Out: analysis_dataset, dq_log

3. Analytics

src/analytics.py

Cuts by income, employment, region, age and loan type. Writes the insights with computed figures.

Tool: pandas

Out: insights.json

4. Model

src/model.py

Logistic regression vs LightGBM, 5 folds grouped by customer. AUC 0.674, Gini 0.347, KS 0.259.

Tools: scikit-learn, LightGBM, SHAP

Out: scores, drivers, model_report

5. Anomaly detection

src/anomaly.py

6 rule detectors and an Isolation Forest. 313 accounts have 2+ signals.

Tools: pandas, scikit-learn

Out: anomaly_flags

6. Dashboard: app/app.py on Streamlit Community Cloud (Streamlit, Plotly). Opens without a login.

Views: Portfolio (filters, risk bands, cross-cuts, trends, insights) | Account (score, drivers, history) | Ask the Portfolio | Anomaly Review | Model & Monitoring | Data Notes

Reads the Parquet and JSON files in data/processed/, which are committed so the hosted app starts instantly. **Writes** analyst decisions and answer feedback to SQLite (src/decisions.py), with CSV export.

AI feature 1: Ask the Portfolio (src/query_engine.py, src/assistant.py)

Step 1. Route. Gemini reads the question and must call query_portfolio, get_account or decline.

Step 2. Compute. Code validates that choice against a fixed list of metrics and dimensions, and pandas computes the result. There is no SQL and no generated code.

Step 3. Phrase. Gemini words the result in 1-3 sentences.

Step 4. Check. Code confirms every number appears in the result. If one does not, a code-written answer replaces it. The source line is always written by code.

Sent to Gemini. Step 1: the question and the catalogue of metric, dimension and value names, with no data rows. Step 3: the question and the computed result (aggregates, or one account's score, band, drivers and flags).

Never sent: customer IDs or raw tables. The API key stays in the host's secrets store.

Without a key, or on a 429: a guided dropdown query uses the same calculations with no LLM.

Gemini API

gemini-3.1-flash-lite

Outside service. 2 calls per question, forced function call on the first. If a model is busy, the next one is tried.

Free tier may use content to improve Google's products: fine for synthetic data, not for production.

AI feature 2: Anomaly Review (src/anomaly.py, src/decisions.py)

No LLM, and no data leaves the app.

Own history: consecutive missed payments, sudden deterioration, clean record then a miss.

Peer segment: far behind peers, high bureau debt for the income band.

Bureau and overall: multiple write-offs, and an Isolation Forest across all signals.

An account joins the ranked review list when 2+ agree. The analyst sees why, with the account's own figures, and records Confirm or Dismiss with a note. Dismiss rates per detector feed threshold reviews.

target_default is never used to fit a detector. As a sanity check only, the list defaults at 32.9% against 16.1% overall.